Motion in a Straight Line

Syllabus Context: Frame of reference, Motion in a straight line (Rectilinear Motion), Position-time graph, Speed and Velocity (average and instantaneous), Uniform and non-uniform motion, Average and instantaneous acceleration, Kinematic equations for uniformly accelerated motion, Free fall, and Relative velocity in one dimension.

1. Learning Objectives & Overview

Motion is common to everything in the universe—from blood flowing in arteries and leaves falling from trees to the Earth revolving around the Sun and galaxies moving within their local groups. In physics, **motion** is defined as the change in position of an object with time.

Key Learning Objectives for NEET UG

- Understand the **point object approximation** and when a physical body can be treated as a particle.
- Distinguish clearly between path length (distance) and displacement, and between speed and velocity over finite intervals versus at a single instant.
- Determine instantaneous velocity and acceleration both analytically (using differential calculus) and graphically (using slopes of tangent lines).
- Calculate displacement from the area under a velocity-time (v-t) curve.
- Derive and apply the kinematic equations of motion for constant acceleration, including vertical motion under gravity (free fall), Galileo's law of odd numbers, stopping distance, and reaction time.
- Analyze one-dimensional relative motion between two moving bodies.

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2. Core Concepts & Explanations

2.1 Kinematics, Rectilinear Motion, and the Point Object Approximation

Kinematics is the branch of mechanics that studies ways to describe motion without going into the causes of motion. When we confine our study to the motion of objects along a single straight line, it is called **rectilinear motion** (or one-dimensional motion).

To specify the position of an object along a straight line, we choose a convenient reference point as the **origin** ($x = 0$). By convention, positions to the right of the origin (or upward) are often taken as positive, and positions to the left (or downward) as negative, though the choice of origin and positive direction of the axis is entirely a matter of convenience.

Point Object Approximation

In kinematics, we treat moving objects as **point objects** (particles). This approximation is valid so far as the size of the object is much smaller than the distance it moves in a reasonable duration of time. For example, a railway carriage moving without jerks between two stations or a monkey sitting on top of a man cycling smoothly on a circular track can be treated as point objects. Conversely, a spinning cricket ball that turns sharply on hitting the ground or a tumbling beaker slipping off the edge of a table cannot be treated as point objects because their size and rotational motion are significant over the short distances considered.

2.2 Path Length vs. Displacement and Average Speed vs. Average Velocity

Consider a particle moving along the x-axis from an initial position x_1 at time t_1 to a final position x_2 at time t_2:

- **Displacement (Δx):** The change in position of the object, given by $\Delta x = x_2 - x_1$. It is an algebraic (vector in 1D) quantity whose sign indicates the direction of net motion.
- **Path Length (Total Distance):** The actual total length of the path traversed by the particle during the time interval $\Delta t = t_2 - t_1$. Path length is always positive and never decreases with time.
- **Average Velocity ($\bar{v}$):** The net displacement divided by the total time interval:
$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t}$$
- **Average Speed:** The total path length divided by the total time interval:
$$\text{Average Speed} = \frac{\text{Total Path Length}}{\text{Total Time Interval}}$$

Because the total path length is always greater than or equal to the magnitude of displacement, **the average speed of an object is always greater than or equal to the magnitude of its average velocity** over a given time interval. The equality holds true *if and only if* the object moves along a straight line in the same constant direction without turning back.

2.3 Instantaneous Velocity and Instantaneous Speed

Average velocity tells us how fast an object has been moving over a finite time interval, but it does not tell us how fast it moves at different instants during that interval. To describe motion at a specific instant t, we define **instantaneous velocity** (or simply **velocity** v) as the limit of the average velocity as the time interval Δt becomes infinitesimally small:

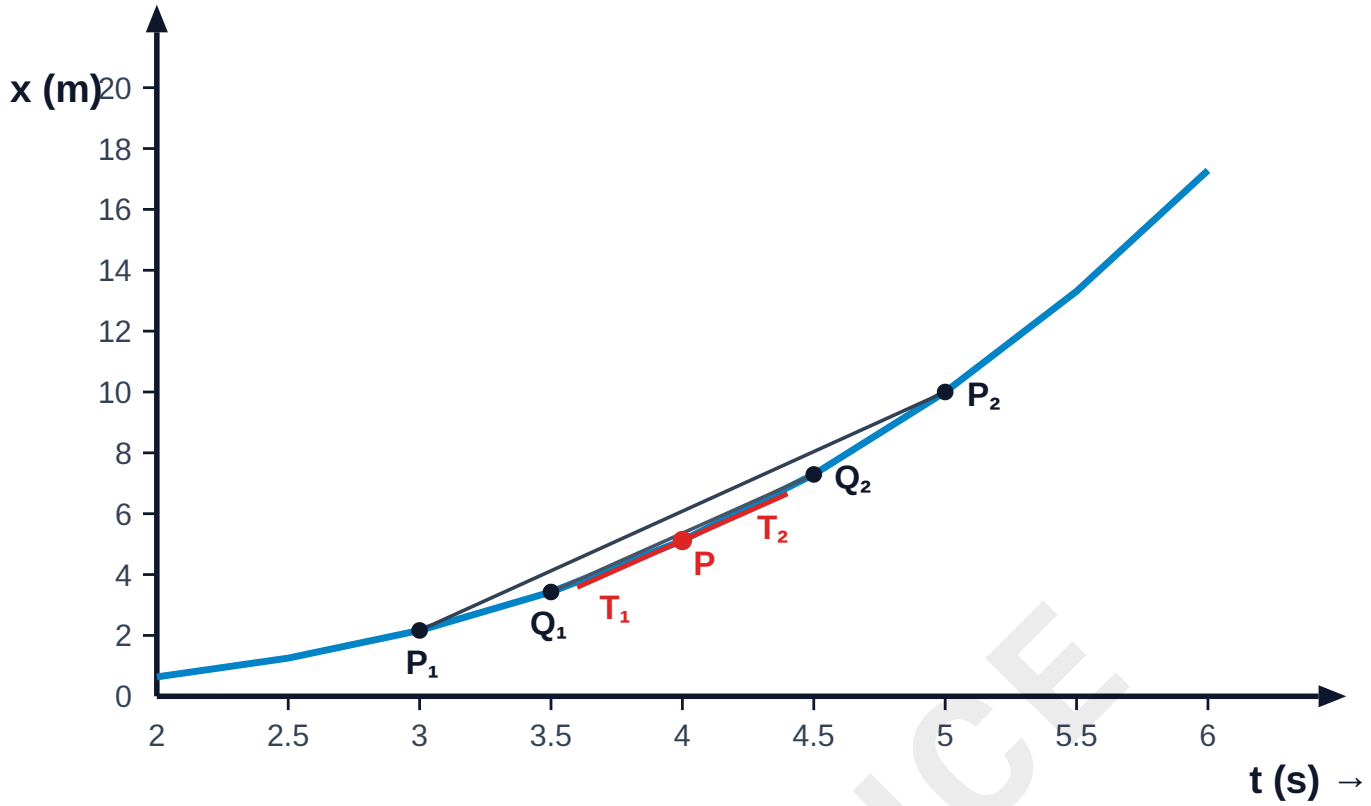
$$v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}$$

In the language of calculus, $\frac{\mathrm{d}x}{\mathrm{d}t}$ is the differential coefficient of position x with respect to time t, representing the instantaneous rate of change of position with respect to time.

Graphical and Numerical Meaning of the Limiting Process

Suppose a car moves along a straight line such that its position as a function of time is given by $x(t) = 0.08\,t^3\ \mathrm{m}$. If we want to find its velocity at $t = 4\,\mathrm{s}$ (marked as point P on the position-time graph below), we can take a time interval Δt centred at $t = 4\,\mathrm{s}$ between $t_1 = 4 - \frac{\Delta t}{2}$ and $t_2 = 4 + \frac{\Delta t}{2}$.

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Tutorial Diagram 1: Determining instantaneous velocity from a position-time (x-t) graph. As Δt approaches zero, secant chords P_1P_2 and Q_1Q_2 approach the tangent T_1T_2 at point P (t = 4 s).

For $\Delta t = 2.0\,\text{s}$, the slope of chord P_1P_2 gives the average velocity between $3.0\,\text{s}$ and $5.0\,\text{s}$. Decreasing Δt to $1.0\,\text{s}$ gives chord Q_1Q_2 over the interval $3.5\,\text{s}$ to $4.5\,\text{s}$. In the limit $\Delta t \to 0$, the chord becomes the tangent line T_1T_2 to the position-time curve at point P, and the instantaneous velocity at $t = 4.0\,\text{s}$ is equal to the **slope of the tangent** at that instant. The numerical convergence of $\frac{\Delta x}{\Delta t}$ for $x(t) = 0.08\,t^3$ centred at $t = 4.0\,\text{s}$ is shown in the table below:

Δt (s)	$t_1 = t - \frac{\Delta t}{2}$ (s)	$t_2 = t + \frac{\Delta t}{2}$ (s)	$x(t_1) = 0.08\,t_1^3$ (m)	$x(t_2) = 0.08\,t_2^3$ (m)	Δx (m)	$\Delta x/\Delta t\,(\text{m s}^{-1})$
2.0	3.0	5.0	2.16	10.0	7.84	3.92
1.0	3.5	4.5	3.43	7.29	3.86	3.86
0.5	3.75	4.25	4.21875	6.14125	1.9225	3.845
0.1	3.95	4.05	4.93039	5.31441	0.38402	3.8402
0.01	3.995	4.005	5.100824	5.139224	0.0384	3.8400

As Δt decreases from $2.0\,\text{s}$ to $0.01\,\text{s}$, the average velocity approaches the limiting value of $3.84\,\text{m s}^{-1}$. Using differential calculus directly:

$$v = \frac{\text{d}x}{\text{d}t} = \frac{\text{d}}{\text{d}t}(0.08\,t^3) = 0.24\,t^2$$

At $t = 4.0\,\text{s}$, $v = 0.24 \times (4.0)^2 = 3.84\,\text{m s}^{-1}$. Note that for **uniform motion** (constant velocity), the instantaneous velocity is the same as the average velocity at all instants.

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Why Instantaneous Speed Always Equals the Magnitude of Instantaneous Velocity

Instantaneous speed (or simply **speed**) is the magnitude of instantaneous velocity: $\text{Speed} = |v| = \left|\frac{\mathrm{d}x}{\mathrm{d}t}\right|$. For example, velocities of $+24.0\,\mathrm{m\,s^{-1}}$ and $-24.0\,\mathrm{m\,s^{-1}}$ both have an associated speed of $24.0\,\mathrm{m\,s^{-1}}$.

Although average speed over a finite time interval can be strictly greater than the magnitude of average velocity (if the particle reverses direction), **at an instant ($\Delta t \to 0$), an infinitesimally small time interval allows no reversal of direction**. Within $\mathrm{d}t$, the particle moves strictly in one direction, so the infinitesimal path length $\mathrm{d}s$ is identical to the magnitude of infinitesimal displacement $|\mathrm{d}x|$. Consequently, instantaneous speed is *always* equal to the magnitude of instantaneous velocity.

2.4 Acceleration

When the velocity of an object changes during its course of motion, how should we describe this change—as the rate of change of velocity with distance or with time? Even in Galileo's time this was a central question. Through his studies of freely falling bodies and motion on inclined planes, **Galileo concluded that the rate of change of velocity with time is a constant of motion for all objects in free fall**, whereas the change in velocity with distance decreases as the distance of fall increases. This led to the definition of **acceleration** as the rate of change of velocity with time.

Average Acceleration ($\bar{a}$) over a time interval $\Delta t = t_2 - t_1$:

$$\bar{a} = \frac{v_2 - v_1}{t_2 - t_1} = \frac{\Delta v}{\Delta t}$$

Instantaneous Acceleration (a) at an instant t:

$$a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}x}$$

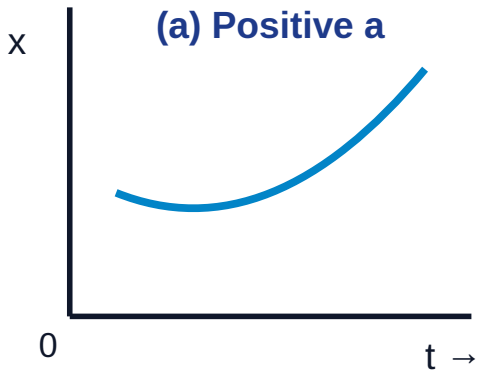
The SI unit of acceleration is $\mathrm{m\,s^{-2}}$ and its dimensional formula is $[\mathrm{L\,T^{-2}}]$. On a velocity-time (v-t) plot, the average acceleration is the slope of the straight line connecting (t_1, v_1) and (t_2, v_2), while the instantaneous acceleration is the **slope of the tangent to the v-t curve** at that instant.

Curvature of Position-Time (x-t) Graphs and Sign of Acceleration

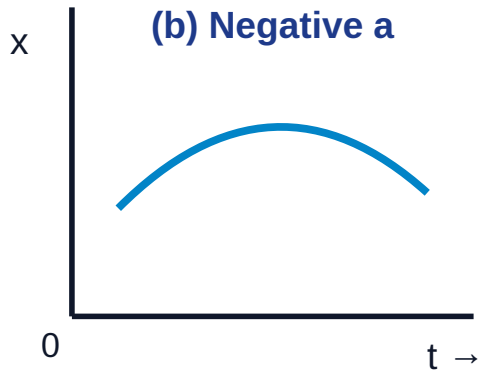
Like velocity, acceleration in one dimension can be positive, negative, or zero:

- **Positive Acceleration ($a > 0$):** The slope of the x-t graph ($\frac{\mathrm{d}x}{\mathrm{d}t}$) increases with time, so the x-t graph **curves upward** (concave upward).
- **Negative Acceleration ($a < 0$):** The slope of the x-t graph decreases with time, so the x-t graph **curves downward** (concave downward).
- **Zero Acceleration ($a = 0$):** Velocity is constant, so the x-t graph is a **straight line**.

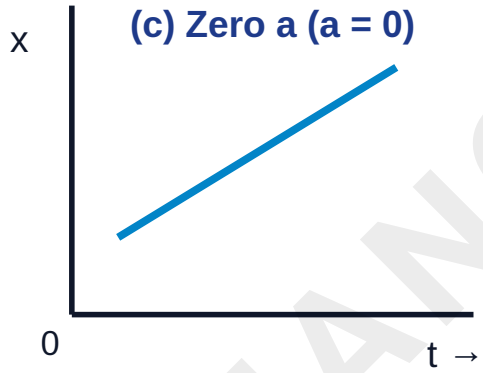
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Curves upward (concave up)



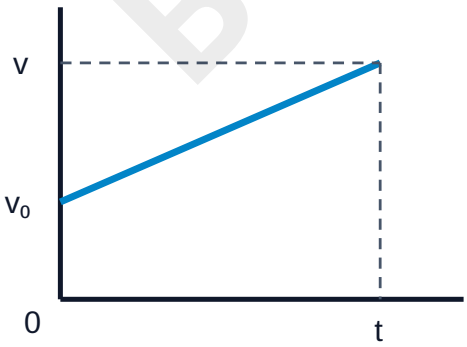
Curves downward (concave down)



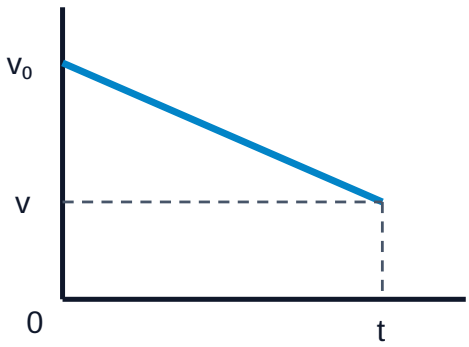
Straight line (uniform motion)

Velocity-Time (v-t) Graphs for Constant Acceleration

When acceleration a is constant, the average acceleration equals the instantaneous acceleration over any interval, and $v = v_0 + at$. Thus, the v-t graph is a straight line inclined to the time axis. The four fundamental cases of constant-acceleration v-t graphs are illustrated below:

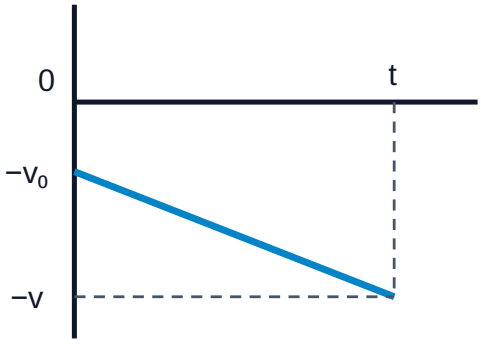


(a) Moving in +x direction with positive acceleration (v > 0, a > 0; speeding up).

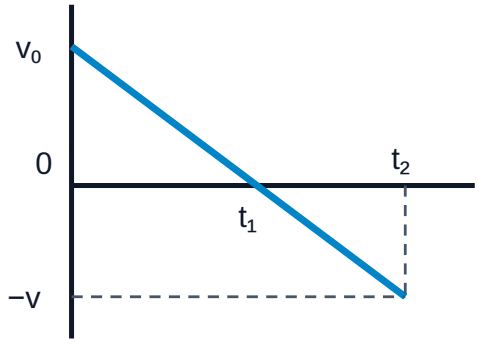


(b) Moving in +x direction with negative acceleration (v > 0, a < 0; slowing down).

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(c) Moving in −x direction with negative acceleration (v < 0, a < 0; speeding up in −x direction).



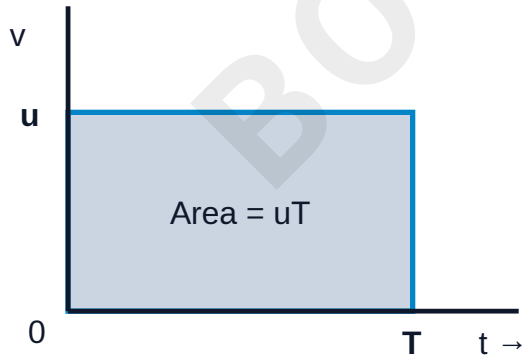
(d) Moving in +x direction until t_1 (where v = 0), then turning back in −x direction under the same negative acceleration.

Note on Realistic Motion vs. Sharp Kinks on Graphs

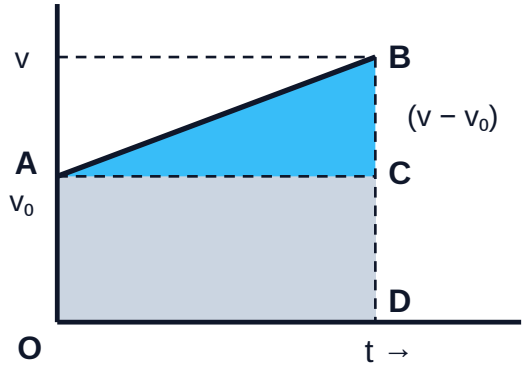
Idealized textbook graphs sometimes show sharp corners or kinks at certain points, which mathematically imply that the functions are not differentiable at those points. In any realistic physical situation, velocity and acceleration cannot change values abruptly at an instant—changes are always continuous, meaning realistic x-t, v-t, and a-t graphs are smooth everywhere.

2.5 Area Under the Velocity-Time Graph & Kinematic Equations

A fundamental property of any velocity-time (v-t) graph is that **the area under the v-t curve between two instants represents the displacement** over that time interval (with areas above the time axis taken as positive and areas below taken as negative; the sum of absolute areas gives the total path length). How can an area on a graph equal a distance? Because the vertical axis has dimensions of velocity ($[\mathrm{L\,T^{-1}}]$) and the horizontal axis has dimensions of time ($[\mathrm{T}]$), so their product has dimensions $[\mathrm{L\,T^{-1}}][\mathrm{T}] = [\mathrm{L}]$, which is displacement.



Tutorial Diagram 2a: Constant velocity u. Area of rectangle = u × T = displacement.



Tutorial Diagram 2b: Uniformly accelerated motion. Total area = Area(OACD) + Area(ABC).

Graphical Derivation of Kinematic Equations for Constant Acceleration

Consider an object moving with constant acceleration a, initial velocity v_0 at $t = 0$, and final velocity v at time t:

1. **Velocity-Time Relation:** From the definition of constant acceleration $\bar{a} = a = \frac{v - v_0}{t - 0}$, we obtain:
$$v = v_0 + at$$

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2. **Position-Time Relation:** From Tutorial Diagram 2b, the displacement x is the area under the v-t graph between 0 and t:

$$x = \text{Area of } \triangle\text{ABC} + \text{Area of rectangle OACD} = \frac{1}{2}(v - v_0)t + v_0 t$$

Since $v - v_0 = at$, substituting this into the above relation gives:

$$x = v_0 t + \frac{1}{2}at^2$$

Notice also that $x = \frac{1}{2}(v - v_0)t + v_0 t = \left(\frac{v+v_0}{2}\right)t = \bar{v}t$, where

$$\bar{v} = \frac{v + v_0}{2} \quad (\text{valid for constant acceleration only})$$

This shows that for constant acceleration, the average velocity over any interval is simply the arithmetic mean of the initial and final velocities.

3. **Velocity-Displacement Relation:** Substituting $t = \frac{v-v_0}{a}$ into $x = \bar{v}t = \left(\frac{v+v_0}{2}\right)t$:

$$x = \left(\frac{v + v_0}{2}\right)\left(\frac{v - v_0}{a}\right) = \frac{v^2 - v_0^2}{2a} \implies v^2 = v_0^2 + 2ax$$

If the initial position of the particle at $t = 0$ is x_0 instead of 0, we replace displacement x with $(x - x_0)$, giving the general kinematic equations:

$$v = v_0 + at$$
$$x = x_0 + v_0 t + \frac{1}{2}at^2$$
$$v^2 = v_0^2 + 2a(x - x_0)$$

2.6 Relative Velocity in One Dimension

Motion is always relative to an observer (frame of reference). Consider two objects A and B moving along the same straight line (x-axis) with uniform velocities v_A and v_B measured relative to the ground:

- The **velocity of object A relative to object B** (v_{AB}) is:
$$v_{\mathrm{AB}} = v_\mathrm{A} - v_\mathrm{B}$$
- The **velocity of object B relative to object A** (v_{BA}) is:
$$v_{\mathrm{BA}} = v_\mathrm{B} - v_\mathrm{A} = -v_{\mathrm{AB}}$$

When applying these formulas, always substitute v_A and v_B with their proper algebraic signs according to the chosen positive axis. Specifically, if A and B move in the **same direction**, the magnitude of their relative velocity is the difference of their speeds ($|v_\mathrm{A} - v_\mathrm{B}|$); if they move in **opposite directions**, the magnitude of their relative velocity is the sum of their speeds ($|v_\mathrm{A}| + |v_\mathrm{B}|$).

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3. Summary of Physical Quantities, Dimensions, and Equations

Physical Quantity	Symbol	Dimensions	SI Unit	Defining Equation & Remarks
Path length (Distance)	s or l	$[\mathrm{L}]$	m	Actual length of path traversed; scalar, always ≥ 0.
Displacement	Δx	$[\mathrm{L}]$	m	$\Delta x = x_2 - x_1$. In 1D, its algebraic sign indicates direction.
Velocity: (a) Average (b) Instantaneous	 $\bar{v}$ v	$[\mathrm{L\,T^{-1}}]$	$\mathrm{m\,s^{-1}}$	$\bar{v} = \frac{\Delta x}{\Delta t}$ $v = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}$ (Slope of tangent to x-t graph).
Speed: (a) Average (b) Instantaneous	 v_{avg} $\lvert v \rvert$	$[\mathrm{L\,T^{-1}}]$	$\mathrm{m\,s^{-1}}$	$\text{Average speed} = \frac{\text{Path length}}{\text{Time interval}} \geq \lvert \bar{v} \rvert$ $\text{Instantaneous speed} = \left\lvert \frac{\mathrm{d}x}{\mathrm{d}t} \right\rvert = \lvert v \rvert$.
Acceleration: (a) Average (b) Instantaneous	 $\bar{a}$ a	$[\mathrm{L\,T^{-2}}]$	$\mathrm{m\,s^{-2}}$	$\bar{a} = \frac{\Delta v}{\Delta t}$ $a = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{\mathrm{d}v}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x}$ (Slope of tangent to v-t graph).

4. Comprehensive Worked Examples

The following worked examples cover every major analytical, graphical, and conceptual application required for NEET UG.

Worked Example 1: Instantaneous vs. Average Velocity using Calculus

Problem: The position of an object moving along the x-axis is given by $x = a + bt^2$, where $a = 8.5\,\mathrm{m}$, $b = 2.5\,\mathrm{m\,s^{-2}}$, and t is measured in seconds. What is its velocity at $t = 0\,\mathrm{s}$ and $t = 2.0\,\mathrm{s}$? What is the average velocity between $t = 2.0\,\mathrm{s}$ and $t = 4.0\,\mathrm{s}$?

Solution:

In the notation of differential calculus, the instantaneous velocity is:

$$v = \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(a + bt^2) = 2bt = 2(2.5)t = 5.0\,t\,\mathrm{m\,s^{-1}}$$

- At $t = 0\,\mathrm{s}$: $v = 5.0(0) = 0\,\mathrm{m\,s^{-1}}$.
- At $t = 2.0\,\mathrm{s}$: $v = 5.0(2.0) = 10\,\mathrm{m\,s^{-1}}$.

The average velocity between $t_1 = 2.0\,\mathrm{s}$ and $t_2 = 4.0\,\mathrm{s}$ is:

$$\bar{v} = \frac{x(4.0) - x(2.0)}{4.0 - 2.0} = \frac{(a + 16b) - (a + 4b)}{2.0} = \frac{12b}{2.0} = 6.0\,b = 6.0 \times 2.5 = 15\,\mathrm{m\,s^{-1}}$$

Check: Since x is quadratic in t, acceleration is constant ($a_{\text{acc}} = 2b = 5.0\,\mathrm{m\,s^{-2}}$). Thus $\bar{v} = \frac{v(2.0) + v(4.0)}{2} = \frac{10 + 20}{2} = 15\,\mathrm{m\,s^{-1}}$.

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Worked Example 2: Derivation of Kinematic Equations via Calculus

Problem: Obtain the three equations of motion for constant acceleration using the method of integral calculus.

Solution:

1. First Equation ($v = v_0 + at$): By definition, $a = \frac{\mathrm{d}v}{\mathrm{d}t} \implies \mathrm{d}v = a\,\mathrm{d}t$. Integrating both sides from initial state $(t = 0, v = v_0)$ to final state (t, v) for constant a:

$$\int_{v_0}^{v} \mathrm{d}v = \int_{0}^{t} a\,\mathrm{d}t = a\int_{0}^{t} \mathrm{d}t \implies v - v_0 = at \implies v = v_0 + at$$

2. Second Equation ($x = x_0 + v_0 t + \frac{1}{2}at^2$): By definition, $v = \frac{\mathrm{d}x}{\mathrm{d}t} \implies \mathrm{d}x = v\,\mathrm{d}t$. Integrating from $(t = 0, x = x_0)$ to (t, x):

$$\int_{x_0}^{x} \mathrm{d}x = \int_{0}^{t} v\,\mathrm{d}t = \int_{0}^{t} (v_0 + at)\,\mathrm{d}t \implies x - x_0 = v_0 t + \frac{1}{2}at^2 \implies x = x_0 + v_0 t + \frac{1}{2}at^2$$

3. Third Equation ($v^2 = v_0^2 + 2a(x - x_0)$): Using the chain rule, $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x} \implies v\,\mathrm{d}v = a\,\mathrm{d}x$. Integrating from (x_0, v_0) to (x, v):

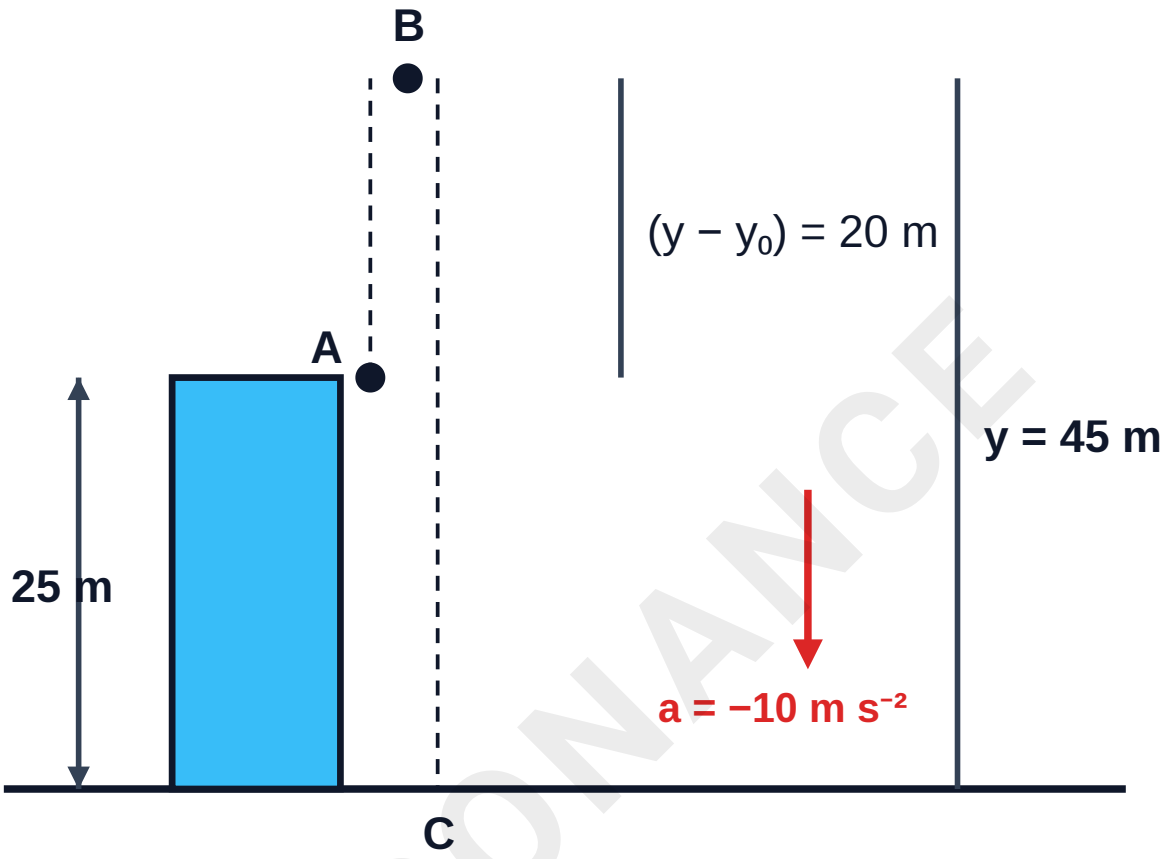
$$\int_{v_0}^{v} v\,\mathrm{d}v = \int_{x_0}^{x} a\,\mathrm{d}x \implies \frac{v^2 - v_0^2}{2} = a(x - x_0) \implies v^2 = v_0^2 + 2a(x - x_0)$$

Key Advantage: Unlike the graphical method, the integral forms $\int \mathrm{d}v = \int a(t)\,\mathrm{d}t$ and $\int v\,\mathrm{d}v = \int a(x)\,\mathrm{d}x$ can be used for **non-uniform acceleration** as well.

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Worked Example 3: Vertical Projection from the Top of a Building

Problem: A ball is thrown vertically upwards with a velocity of $20\,\mathrm{m\,s^{-1}}$ from the top of a multistorey building. The height of the point from where the ball is thrown is $25.0\,\mathrm{m}$ from the ground. (a) How high will the ball rise? (b) How long will it be before the ball hits the ground? Take $g = 10\,\mathrm{m\,s^{-2}}$.



Tutorial Diagram 3: Vertical projection of a ball from a 25 m building (A) to highest point (B) and ground (C).

Solution:

Take the y-axis in the vertically upward direction with the origin ($y = 0$) at the ground.

(a) Maximum height risen above the launch point:
Given $v_0 = +20\,\mathrm{m\,s^{-1}}$, $a = -g = -10\,\mathrm{m\,s^{-2}}$, and at the highest point B, $v = 0\,\mathrm{m\,s^{-1}}$. Using $v^2 = v_0^2 + 2a(y - y_0)$:

$$0 = (20)^2 + 2(-10)(y - y_0) \implies y - y_0 = \frac{400}{20} = 20\,\mathrm{m}$$

Thus, the ball rises $20\,\mathrm{m}$ above the top of the building (reaching a height $y = 25 + 20 = 45\,\mathrm{m}$ above the ground).

(b) Total time before hitting the ground:

- **First Method (Splitting the path into upward A→B and downward B→C):**
  For upward motion $\mathrm{A} \to \mathrm{B}$, $v = v_0 + at_1 \implies 0 = 20 - 10t_1 \implies t_1 = 2\,\mathrm{s}$.
  From maximum height B ($y_0 = 45\,\mathrm{m}$, $v_0 = 0$) to ground C ($y = 0$):
  $$y = y_0 + v_0 t_2 + \frac{1}{2}at_2^2 \implies 0 = 45 + 0 + \frac{1}{2}(-10)t_2^2 \implies 5t_2^2 = 45 \implies t_2 = 3\,\mathrm{s}$$

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Total time $t = t_1 + t_2 = 2\,\text{s} + 3\,\text{s} = 5\,\text{s}$.

- **Second Method (Direct algebraic displacement from A to C — Recommended for NEET):** Since acceleration is constant throughout the entire flight, apply $y = y_0 + v_0 t + \frac{1}{2}at^2$ directly between initial position $y_0 = 25\,\text{m}$ and final ground position $y = 0\,\text{m}$:

$$0 = 25 + 20t + \frac{1}{2}(-10)t^2 \implies 5t^2 - 20t - 25 = 0 \implies t^2 - 4t - 5 = 0$$

$$(t-5)(t+1) = 0 \implies t = 5\,\text{s} \quad (\text{since } t > 0)$$

The second method is much faster because we do not need to split the path when acceleration is constant.

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Worked Example 4: Motion of an Object Under Free Fall & Graphical Representation

Problem: Discuss the motion of an object released from rest under free fall near the Earth's surface (neglecting air resistance) and plot the variations of acceleration, velocity, and position with time.

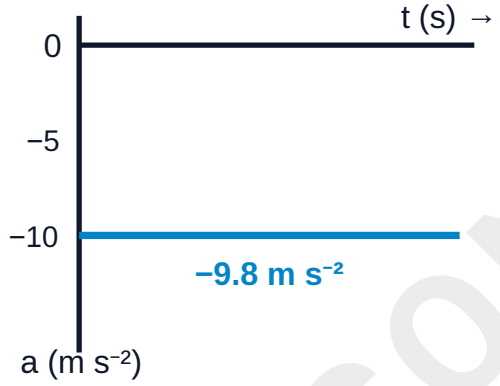
Solution:

An object released near the Earth's surface is accelerated downward under gravity with magnitude $g = 9.8\,\mathrm{m\,s^{-2}}$. If air resistance is neglected and the height of fall is small compared to the Earth's radius, the motion is called **free fall** and has constant acceleration. Choosing the upward direction as positive $+y$ and the release point as $y = 0$ with $v_0 = 0$, we have $a = -g = -9.8\,\mathrm{m\,s^{-2}}$, and the kinematic equations become:

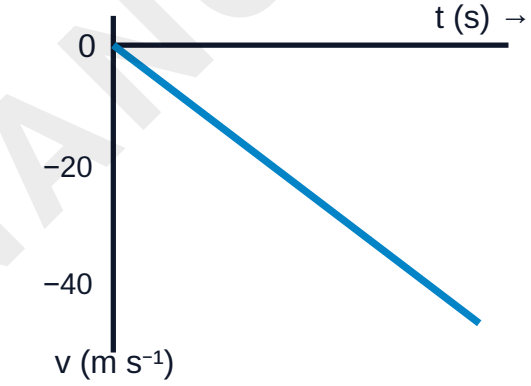
$$v = -gt = -9.8\,t\,\mathrm{m\,s^{-1}}$$

$$y = -\frac{1}{2}gt^2 = -4.9\,t^2\,\mathrm{m}$$

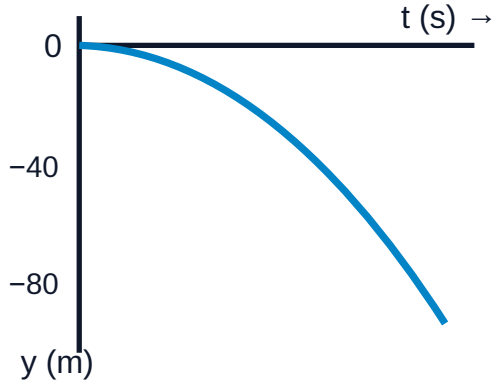
$$v^2 = -2gy = -19.6\,y\,\mathrm{m^2\,s^{-2}}$$



(a) Acceleration vs. time



(b) Velocity vs. time



(c) Position vs. time

BOSONANCE

Worked Example 5: Galileo's Law of Odd Numbers

Problem: Prove Galileo's law of odd numbers: *"The distances traversed, during equal intervals of time, by a body falling from rest, stand to one another in the same ratio as the odd numbers beginning with unity [namely, $1:3:5:7:9:\ldots$]."*

Solution:

Divide the time of free fall from rest ($v_0 = 0$) into equal successive intervals of duration τ. Using $y = -\frac{1}{2}gt^2$ and defining $y_0 = -\frac{1}{2}g\tau^2$ as the position after the first time interval τ, we obtain the positions and successive distances at $t = 0, \tau, 2\tau, 3\tau, \ldots$ as tabulated below:

Time t	Position $y = -\frac{1}{2}gt^2$	y in terms of $y_0 = -\frac{1}{2}g\tau^2$	Distance traversed in successive τ	Ratio of successive distances
0	0	0	–	–
τ	$-\frac{1}{2}g\tau^2$	y_0	y_0	1
2τ	$-4\left(\frac{1}{2}g\tau^2\right)$	$4y_0$	$4y_0 - y_0 = 3y_0$	3
3τ	$-9\left(\frac{1}{2}g\tau^2\right)$	$9y_0$	$9y_0 - 4y_0 = 5y_0$	5
4τ	$-16\left(\frac{1}{2}g\tau^2\right)$	$16y_0$	$16y_0 - 9y_0 = 7y_0$	7
5τ	$-25\left(\frac{1}{2}g\tau^2\right)$	$25y_0$	$25y_0 - 16y_0 = 9y_0$	9
6τ	$-36\left(\frac{1}{2}g\tau^2\right)$	$36y_0$	$36y_0 - 25y_0 = 11y_0$	11

In general, the distance travelled in the nth equal time interval τ from rest is $\Delta y_n = n^2 y_0 - (n-1)^2 y_0 = (2n-1)y_0$, giving the exact ratio $1:3:5:7:9:11:\ldots$.

BOSONANCE

Worked Example 6: Stopping Distance of Vehicles & Highway Retardation

Problem: (a) Derive an expression for the stopping distance d_s of a vehicle moving with initial velocity v_0 when brakes apply a uniform deceleration a (where $a < 0$). (b) A car moving along a straight highway with a speed of $126\,\mathrm{km\,h^{-1}}$ is brought to a stop within a distance of $200\,\mathrm{m}$. What is the uniform retardation of the car, and how long does it take for the car to stop?

Solution:

(a) Stopping Distance Formula: When the vehicle comes to a complete stop, its final velocity is $v = 0$. Using $v^2 = v_0^2 + 2ax$ with $x = d_s$:

$$0 = v_0^2 + 2ad_s \implies d_s = \frac{-v_0^2}{2a}$$

Since $d_s \propto v_0^2$ for a given deceleration, **doubling the initial velocity increases the stopping distance by a factor of** 4, and tripling the initial velocity increases it by a factor of 9. (For instance, experimental braking distances of $10\,\mathrm{m}$, $20\,\mathrm{m}$, $34\,\mathrm{m}$, and $50\,\mathrm{m}$ for speeds of $11\,\mathrm{m\,s^{-1}}$, $15\,\mathrm{m\,s^{-1}}$, $20\,\mathrm{m\,s^{-1}}$, and $25\,\mathrm{m\,s^{-1}}$ closely follow $d_s \propto v_0^2$).

(b) Highway Car Calculation:
Convert initial speed to SI units: $v_0 = 126\,\mathrm{km\,h^{-1}} = 126 \times \frac{5}{18}\,\mathrm{m\,s^{-1}} = 35\,\mathrm{m\,s^{-1}}$.
Given stopping distance $d_s = 200\,\mathrm{m}$ and $v = 0$:

$$a = \frac{-v_0^2}{2d_s} = \frac{-(35)^2}{2 \times 200} = \frac{-1225}{400} = -3.0625\,\mathrm{m\,s^{-2}} \approx -3.06\,\mathrm{m\,s^{-2}}$$

Thus, the uniform retardation is $3.06\,\mathrm{m\,s^{-2}}$. The time taken to stop is found from $v = v_0 + at$:

$$t = \frac{0 - v_0}{a} = \frac{-35}{-3.0625} = \frac{80}{7}\,\mathrm{s} \approx 11.43\,\mathrm{s}$$

Worked Example 7: Reaction Time Measurement

Problem: Reaction time is the time a person takes to observe, think, and act. In a simple experiment, a friend holds a ruler vertically above the gap between your thumb and forefinger and drops it without warning. You catch it after it has fallen a vertical distance $d = 21.0\,\mathrm{cm}$. Estimate your reaction time t_r (take $g = 9.8\,\mathrm{m\,s^{-2}}$).

Solution:

The ruler drops from rest ($v_0 = 0$) under free fall ($a = -g = -9.8\,\mathrm{m\,s^{-2}}$). Taking the downward distance fallen as $d = 21.0\,\mathrm{cm} = 0.21\,\mathrm{m}$, the magnitude of displacement is:

$$d = \frac{1}{2}gt_r^2 \implies t_r = \sqrt{\frac{2d}{g}} = \sqrt{\frac{2 \times 0.21}{9.8}} = \sqrt{\frac{0.42}{9.8}} = \sqrt{\frac{3}{70}}\,\mathrm{s} \approx 0.207\,\mathrm{s} \approx 0.2\,\mathrm{s}$$

BOSONANCE

Worked Example 8: Average Velocity vs. Average Speed in Round-Trip Motion

Problem: A man walks on a straight road from his home to a market $2.5\,\text{km}$ away with a speed of $5\,\text{km}\,\text{h}^{-1}$. Finding the market closed, he instantly turns and walks back home with a speed of $7.5\,\text{km}\,\text{h}^{-1}$. What is the (a) magnitude of average velocity, and (b) average speed of the man over the time interval: (i) 0 to $30\,\text{min}$, (ii) 0 to $50\,\text{min}$, and (iii) 0 to $40\,\text{min}$?

Solution:

Time taken to reach the market ($2.5\,\text{km}$ at $5\,\text{km}\,\text{h}^{-1}$): $t_1 = \frac{2.5\,\text{km}}{5\,\text{km}\,\text{h}^{-1}} = 0.5\,\text{h} = 30\,\text{min}$.
Time taken to walk back home ($2.5\,\text{km}$ at $7.5\,\text{km}\,\text{h}^{-1}$): $t_2 = \frac{2.5\,\text{km}}{7.5\,\text{km}\,\text{h}^{-1}} = \frac{1}{3}\,\text{h} = 20\,\text{min}$.
Total time for the round trip $= 30 + 20 = 50\,\text{min} = \frac{5}{6}\,\text{h}$.

- **(i) Interval 0 to $30\,\text{min}$ ($0.5\,\text{h}$):** The man walks straight from home to the market without turning.
  Displacement $= 2.5\,\text{km}$, Path length $= 2.5\,\text{km}$.
  Magnitude of average velocity $= \frac{2.5\,\text{km}}{0.5\,\text{h}} = 5\,\text{km}\,\text{h}^{-1}$.
  Average speed $= \frac{2.5\,\text{km}}{0.5\,\text{h}} = 5\,\text{km}\,\text{h}^{-1}$.
- **(ii) Interval 0 to $50\,\text{min}$ ($\frac{5}{6}\,\text{h}$):** The man returns to his starting point (home).
  Net displacement $= 0\,\text{km}$, Total path length $= 2.5 + 2.5 = 5.0\,\text{km}$.
  Magnitude of average velocity $= \frac{0}{5/6\,\text{h}} = 0\,\text{km}\,\text{h}^{-1}$.
  Average speed $= \frac{5.0\,\text{km}}{5/6\,\text{h}} = 6.0\,\text{km}\,\text{h}^{-1}$.
- **(iii) Interval 0 to $40\,\text{min}$ ($\frac{2}{3}\,\text{h}$):** In the first $30\,\text{min}$, he covers $2.5\,\text{km}$ forward. In the next $10\,\text{min}$ ($\frac{1}{6}\,\text{h}$), he walks back toward home a distance of $7.5\,\text{km}\,\text{h}^{-1} \times \frac{1}{6}\,\text{h} = 1.25\,\text{km}$.
  Net displacement $= 2.5 - 1.25 = 1.25\,\text{km}$; Total path length $= 2.5 + 1.25 = 3.75\,\text{km}$.
  Magnitude of average velocity $= \frac{1.25\,\text{km}}{2/3\,\text{h}} = 1.875\,\text{km}\,\text{h}^{-1}$.
  Average speed $= \frac{3.75\,\text{km}}{2/3\,\text{h}} = 5.625\,\text{km}\,\text{h}^{-1}$.

BOSONANCE

Worked Example 9: Forward-and-Backward Step Motion (The Drunkard Problem)

Problem: A drunkard walking in a narrow lane takes 5 steps forward and 3 steps backward, followed again by 5 steps forward and 3 steps backward, and so on. Each step is $1\,\mathrm{m}$ long and requires $1\,\mathrm{s}$. Determine how long the drunkard takes to fall into a pit $13\,\mathrm{m}$ away from the start.

Solution:

In one complete cycle of 5 steps forward and 3 steps backward:

- Time taken per cycle $= 5\,\mathrm{s} + 3\,\mathrm{s} = 8\,\mathrm{s}$.
- Net forward displacement per cycle $= 5\,\mathrm{m} - 3\,\mathrm{m} = 2\,\mathrm{m}$.

Crucial Trap to Avoid: Once the drunkard reaches $x = 13\,\mathrm{m}$, he falls into the pit and does *not* take backward steps! Since his final forward run covers $5\,\mathrm{m}$, we must first find the time taken to reach $13\,\mathrm{m} - 5\,\mathrm{m} = 8\,\mathrm{m}$.

- Number of cycles to reach $x = 8\,\mathrm{m}$ is $\frac{8\,\mathrm{m}}{2\,\mathrm{m/cycle}} = 4\ \text{cycles}$.
- Time for 4 cycles $= 4 \times 8\,\mathrm{s} = 32\,\mathrm{s}$ (at $t = 32\,\mathrm{s}$, he is at $x = 8\,\mathrm{m}$).
- Next, he takes 5 steps forward ($5\,\mathrm{m}$ in $5\,\mathrm{s}$), reaching $x = 8\,\mathrm{m} + 5\,\mathrm{m} = 13\,\mathrm{m}$ at time:

$$t = 32\,\mathrm{s} + 5\,\mathrm{s} = 37\,\mathrm{s}$$

Thus, the drunkard falls into the pit at $t = 37\,\mathrm{s}$.

Worked Example 10: Relative Velocity of a Bullet Hitting a Moving Car

Problem: A police van moving on a highway with a speed of $30\,\mathrm{km\,h^{-1}}$ fires a bullet at a thief's car speeding away in the same direction with a speed of $192\,\mathrm{km\,h^{-1}}$. If the muzzle speed of the bullet (relative to the police van) is $150\,\mathrm{m\,s^{-1}}$, with what speed does the bullet hit the thief's car?

Solution:

Convert all vehicle speeds to $\mathrm{m\,s^{-1}}$ (multiplying $\mathrm{km\,h^{-1}}$ by $\frac{5}{18}$):

- Speed of police van relative to ground: $v_{\mathrm{P}} = 30 \times \frac{5}{18} = \frac{25}{3}\,\mathrm{m\,s^{-1}}$.
- Speed of thief's car relative to ground: $v_{\mathrm{T}} = 192 \times \frac{5}{18} = \frac{160}{3}\,\mathrm{m\,s^{-1}}$.
- Muzzle speed of bullet relative to police van: $v_{\mathrm{BP}} = 150\,\mathrm{m\,s^{-1}}$.

Since the bullet is fired in the forward direction from the moving police van, the speed of the bullet relative to the ground is:

$$v_{\mathrm{B}} = v_{\mathrm{BP}} + v_{\mathrm{P}} = 150 + \frac{25}{3} = \frac{475}{3}\,\mathrm{m\,s^{-1}}$$

The speed relevant for damaging the thief's car is the **relative speed of the bullet with respect to the thief's car** (v_{BT}):

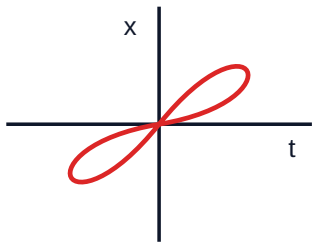
$$v_{\mathrm{BT}} = v_{\mathrm{B}} - v_{\mathrm{T}} = \frac{475}{3} - \frac{160}{3} = \frac{315}{3} = 105\,\mathrm{m\,s^{-1}}$$

BOSONANCE

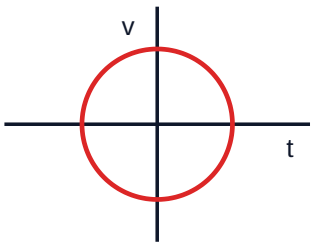
Worked Example 11: Interpreting Kinematics Graphs (Physical Validity, Slope, and Curvature)

Problem: Analyze the following three sets of graphical scenarios fundamental to one-dimensional motion:

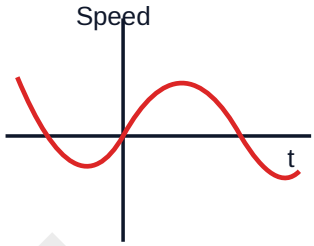
Part I: Examine the four graphs (a) to (d) below and explain why *none* of them can possibly represent the one-dimensional motion of a physical particle.



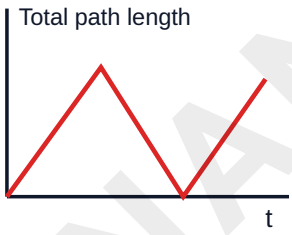
(a) Looped x-t curve



(b) Circular v-t curve

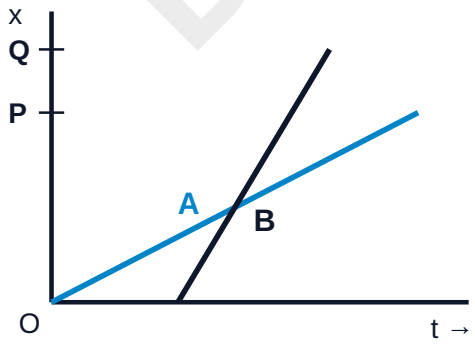


(c) Speed-t crossing below 0

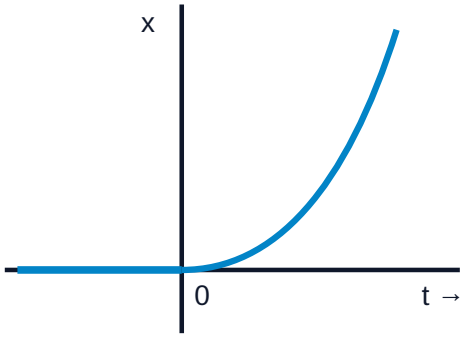


(d) Zig-zag path length vs. t

Part II: In Tutorial Diagram 4a below, two children A and B return from their school O to their homes P and Q. Compare their home distances, start times, walking speeds, arrival times, and overtaking events. In Tutorial Diagram 4b, a particle's x-t graph is horizontal along $x = 0$ for $t < 0$ and curves parabolically upward for $t > 0$. Is it correct to say that the particle moves in a straight line for $t < 0$ and on a parabolic path in space for $t > 0$?



Tutorial Diagram 4a: x-t graphs of children A and B.



Tutorial Diagram 4b: x-t graph for t < 0 and t > 0.

Solution:

Part I (Why graphs a–d are impossible):

BOSONANCE

- **Graph (a):** A vertical line at a single instant t intersects the x-t curve at two points, implying the particle occupies two different positions at the same time—physically impossible.
- **Graph (b):** A vertical line at time t intersects the v-t circle at two points, implying the particle has two different velocities (in opposite directions) at the exact same instant—impossible.
- **Graph (c):** Speed is the magnitude of velocity ($|v| \geq 0$) and can *never* be negative.
- **Graph (d):** Total path length is the cumulative distance travelled and can *never decrease* with increasing time.

Part II (Tutorial Diagrams 4a and 4b):

- **Children A and B:** (1) Since $\mathrm{OP} < \mathrm{OQ}$, A **lives closer** to the school than B. (2) A starts at $t = 0$ while B starts at a later positive time, so A **starts earlier** than B. (3) The slope of B's x-t line is steeper than A's, so B **walks faster** than A. (4) Their endpoints correspond to different time coordinates, so they reach home at **different times**. (5) The two lines intersect at exactly one point, so B **overtakes** A **on the road once**.
- **Trajectory vs. x-t Graph (Diagram 4b): No!** An x-t graph is *not* the spatial trajectory of the particle. For $t < 0$, $x = 0$ is constant, meaning the particle is **at rest** at the origin (not moving!). For $t > 0$, it moves along a **straight line** (x-axis) with constant positive acceleration ($x = \frac{1}{2}at^2$), such as a body released from rest at $t = 0$ under a constant force (e.g., free fall).

5. NEET UG Exam-Aligned Practice Questions

Note: All questions below are original practice questions designed strictly in accordance with the NEET UG syllabus and examination pattern (Single-Correct MCQs, Multi-Statement Evaluations, Assertion-Reason, and Matrix-Match).

Practice Question 1 [Conceptual — Point Object Approximation]

In which of the following situations can the moving body **NOT** be approximated as a point object?

(A) A railway carriage moving without jerks between two cities separated by $150\ \mathrm{km}$.

(B) A monkey sitting on top of a man cycling smoothly on a large circular track.

(C) A spinning cricket ball that turns sharply upon hitting the pitch.

(D) An aeroplane flying between two airports separated by $800\ \mathrm{km}$.

Correct Answer: (C)
Worked Solution: A body can be treated as a point object only when its dimensions are negligible compared to the distance it moves in a reasonable time interval, and its rotational/tumbling motion does not dominate the phenomenon. A spinning cricket ball turning sharply on hitting the ground undergoes significant rotational and surface-contact effects over a distance comparable to its own size, so it cannot be treated as a point object.

BOSONANCE

Practice Question 2 [Differential Calculus — Instantaneous Velocity & Acceleration]

The position x (in metres) of a particle moving along a straight line varies with time t (in seconds) as $x(t) = 2t^3 - 12t^2 + 18t + 5$. What is the acceleration of the particle at the instants when its instantaneous velocity becomes zero?

(A) $0\,\mathrm{m\,s^{-2}}$ and $+12\,\mathrm{m\,s^{-2}}$

(B) $-12\,\mathrm{m\,s^{-2}}$ and $+12\,\mathrm{m\,s^{-2}}$

(C) $-6\,\mathrm{m\,s^{-2}}$ and $+6\,\mathrm{m\,s^{-2}}$

(D) $-24\,\mathrm{m\,s^{-2}}$ and $+24\,\mathrm{m\,s^{-2}}$

Correct Answer: (B)
Worked Solution: Differentiating $x(t)$ with respect to t gives instantaneous velocity:

$$v(t) = \frac{\mathrm{d}x}{\mathrm{d}t} = 6t^2 - 24t + 18 = 6(t^2 - 4t + 3) = 6(t-1)(t-3)\,\mathrm{m\,s^{-1}}$$

Setting $v(t) = 0$ gives $t = 1\,\mathrm{s}$ and $t = 3\,\mathrm{s}$. Differentiating $v(t)$ gives instantaneous acceleration:

$$a(t) = \frac{\mathrm{d}v}{\mathrm{d}t} = 12t - 24\,\mathrm{m\,s^{-2}}$$

At $t = 1\,\mathrm{s}$, $a(1) = 12(1) - 24 = -12\,\mathrm{m\,s^{-2}}$. At $t = 3\,\mathrm{s}$, $a(3) = 12(3) - 24 = +12\,\mathrm{m\,s^{-2}}$. Notice that even though velocity is momentarily zero at $t = 1\,\mathrm{s}$ and $t = 3\,\mathrm{s}$, acceleration is non-zero!

Practice Question 3 [Chain Rule $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$ — Velocity as a Function of Position]

A particle moves along the positive x-axis such that its time coordinate t varies with position x according to the relation $t = \alpha x^2 + \beta x$, where α and β are positive constants. If v is the instantaneous velocity of the particle, its acceleration a is given by:

(A) $-2\alpha v^3$

(B) $-2\alpha v^2$

(C) $-2\beta v^3$

(D) $+\alpha v^3$

Correct Answer: (A)
Worked Solution: Differentiate $t = \alpha x^2 + \beta x$ with respect to x:

$$\frac{\mathrm{d}t}{\mathrm{d}x} = 2\alpha x + \beta \implies \frac{1}{v} = 2\alpha x + \beta \implies v = (2\alpha x + \beta)^{-1}$$

Now use $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$\frac{\mathrm{d}v}{\mathrm{d}x} = -(2\alpha x + \beta)^{-2}(2\alpha) = -2\alpha v^2$$

Therefore, $a = v(-2\alpha v^2) = -2\alpha v^3$.

BOSONANCE

Practice Question 4 [Average Speed vs. Average Velocity]

A cyclist travels along a straight highway from town A to town B with a constant speed of $20\,\mathrm{km\,h^{-1}}$, and immediately returns from town B to town A with a constant speed of $30\,\mathrm{km\,h^{-1}}$. For the entire round trip, the average speed and the magnitude of average velocity of the cyclist are, respectively:

(A) $25\,\mathrm{km\,h^{-1}}$ and $25\,\mathrm{km\,h^{-1}}$

(B) $24\,\mathrm{km\,h^{-1}}$ and $0\,\mathrm{km\,h^{-1}}$

(C) $0\,\mathrm{km\,h^{-1}}$ and $24\,\mathrm{km\,h^{-1}}$

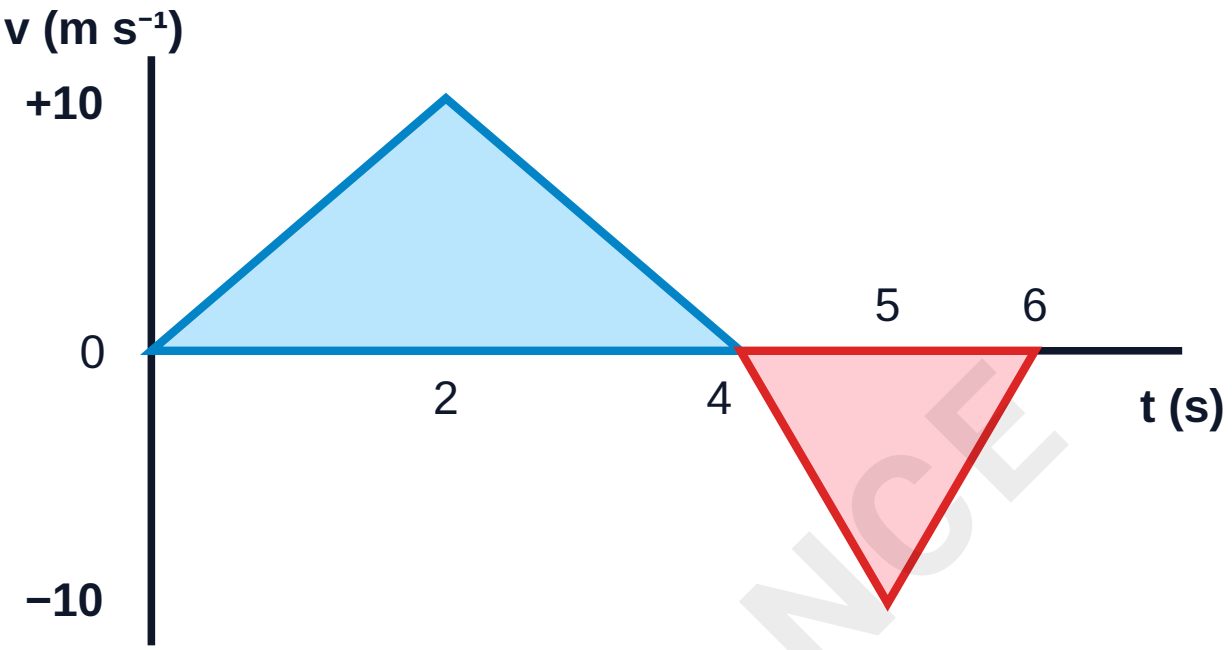
(D) $25\,\mathrm{km\,h^{-1}}$ and $0\,\mathrm{km\,h^{-1}}$

Correct Answer: (B)
Worked Solution: Let the distance between A and B be d. Since the cyclist returns to the starting point A, the net displacement is $\Delta x = 0$, so the magnitude of average velocity is $|\bar{v}| = 0\,\mathrm{km\,h^{-1}}$.
Total path length $= 2d$, and total time $= \frac{d}{20} + \frac{d}{30} = \frac{5d}{60} = \frac{d}{12}$.
Thus, average speed $= \frac{2d}{d/12} = \frac{2\times 20\times 30}{20+30} = 24\,\mathrm{km\,h^{-1}}$.

BOSONANCE

Practice Question 5 [Area Under v-t Curve — Displacement vs. Distance]

A particle moves along a straight line according to the velocity-time (v-t) graph shown below. What is the ratio of the **total path length (distance)** covered to the **magnitude of net displacement** of the particle from $t = 0\,\text{s}$ to $t = 6\,\text{s}$?



(A) $1 : 1$

(B) $3 : 1$

(C) $2 : 1$

(D) $5 : 3$

Correct Answer: (B)
Worked Solution:
Area above the time axis from $t = 0$ to $t = 4\,\text{s}$: $A_1 = \frac{1}{2} \times 4\,\text{s} \times 10\,\text{m s}^{-1} = 20\,\text{m}$.
Area below the time axis from $t = 4\,\text{s}$ to $t = 6\,\text{s}$: $A_2 = \frac{1}{2} \times (6 - 4)\,\text{s} \times 10\,\text{m s}^{-1} = 10\,\text{m}$.
Total path length (distance) $= A_1 + A_2 = 20 + 10 = 30\,\text{m}$.
Magnitude of net displacement $= |A_1 - A_2| = |20 - 10| = 10\,\text{m}$.
Required ratio $= \frac{30\,\text{m}}{10\,\text{m}} = 3 : 1$.

BOSONANCE

Practice Question 6 [Galileo's Law of Odd Numbers — Free Fall]

A stone is dropped from rest from the top of a tall tower. Neglecting air resistance, the ratio of the distance covered by the stone during the 3rd second of its fall to the distance covered during the first 3 seconds of its fall is:

(A) $5 : 9$

(B) $1 : 3$

(C) $3 : 5$

(D) $5 : 16$

Correct Answer: (A)
Worked Solution: By Galileo's law of odd numbers, the distances covered in successive $1\,\text{s}$ intervals from rest are in the ratio $1 : 3 : 5 : 7 : \ldots$ (in units of $y_0 = \frac{1}{2}g(1)^2$).
Distance in the 3rd second $= 5y_0$.
Total distance in the first 3 seconds $= (1 + 3 + 5)y_0 = 9y_0$.
Therefore, the required ratio is $5 : 9$.

Practice Question 7 [Stopping Distance & Reaction Time]

A driver is driving a car at a constant speed of $20\,\text{m}\,\text{s}^{-1}$. Suddenly a child appears on the road ahead. The driver's reaction time is $0.5\,\text{s}$, after which the brakes are applied to produce a uniform retardation of $5.0\,\text{m}\,\text{s}^{-2}$. What is the total distance travelled by the car from the instant the driver sees the child until the car comes to a complete stop?

(A) $40\,\text{m}$

(B) $45\,\text{m}$

(C) $50\,\text{m}$

(D) $60\,\text{m}$

Correct Answer: (C)
Worked Solution:
1. During the reaction time $t_r = 0.5\,\text{s}$, the car continues to move at constant speed $v_0 = 20\,\text{m}\,\text{s}^{-1}$:

$$d_{\text{reaction}} = v_0 t_r = 20 \times 0.5 = 10\,\text{m}$$

2. After the brakes are applied ($a = -5.0\,\text{m}\,\text{s}^{-2}$), the braking distance is:

$$d_s = \frac{-v_0^2}{2a} = \frac{-(20)^2}{2(-5.0)} = \frac{400}{10} = 40\,\text{m}$$

Total stopping distance $= d_{\text{reaction}} + d_s = 10\,\text{m} + 40\,\text{m} = 50\,\text{m}$.

BOSONANCE

Practice Question 8 [Vertical Motion Under Gravity — Coordinate Sign Conventions]

A player throws a ball vertically upwards with an initial speed of $29.4\,\mathrm{m\,s^{-1}}$ (take $g = 9.8\,\mathrm{m\,s^{-2}}$). Choose $x = 0\,\mathrm{m}$ and $t = 0\,\mathrm{s}$ at the highest point of its trajectory, and choose the **vertically downward direction as the positive x-axis**. Which of the following correctly gives the algebraic signs of position (x), velocity (v), and acceleration (a) during the ball's **upward motion** and **downward motion**, as well as the maximum height risen?

(A) Upward: $x > 0, v < 0, a > 0$; Downward: $x > 0, v > 0, a > 0$; Height $= 44.1\,\mathrm{m}$

(B) Upward: $x < 0, v < 0, a < 0$; Downward: $x > 0, v > 0, a > 0$; Height $= 44.1\,\mathrm{m}$

(C) Upward: $x > 0, v < 0, a < 0$; Downward: $x > 0, v > 0, a > 0$; Height $= 29.4\,\mathrm{m}$

(D) Upward: $x < 0, v > 0, a > 0$; Downward: $x > 0, v > 0, a > 0$; Height $= 44.1\,\mathrm{m}$

Correct Answer: (A)
Worked Solution:
Since the origin $x = 0$ is at the highest point and the downward direction is positive ($+x$):

- **Position (x):** Both during upward and downward motion, the ball is below the highest point, so $x > 0$ in both phases.
- **Velocity (v):** During upward motion, the ball moves opposite to the downward $+x$-axis, so $v < 0$. During downward motion, it moves along $+x$, so $v > 0$.
- **Acceleration (a):** Gravity always acts vertically downward (along $+x$), so $a = +g > 0$ during both upward and downward motion!
- **Maximum height:** $H = \frac{v_0^2}{2g} = \frac{(29.4)^2}{2\times 9.8} = \frac{29.4\times 29.4}{19.6} = 1.5 \times 29.4 = 44.1\,\mathrm{m}$ (and total time to return to hands $= \frac{2v_0}{g} = 6.0\,\mathrm{s}$).

BOSONANCE

Practice Question 9 [Multi-Statement Evaluation — Kinematics Fundamentals]

Read the following four statements regarding a particle in one-dimensional motion carefully:

1. **Statement I:** A particle with zero speed at an instant may have non-zero acceleration at that instant.
2. **Statement II:** A particle with zero speed at an instant may have non-zero velocity at that instant.
3. **Statement III:** A particle moving with constant speed in one dimension must have zero acceleration.
4. **Statement IV:** A particle with a positive value of acceleration ($a > 0$) must be speeding up.

Which of the above statements are **TRUE**?

(A) Statements I and III only

(B) Statements I, III, and IV only

(C) Statements I and IV only

(D) Statements II and IV only

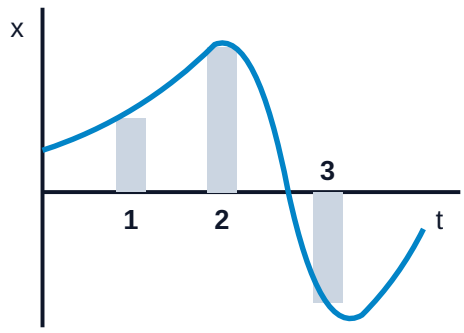
Correct Answer: (A)
Worked Solution:

- **Statement I is TRUE:** A ball thrown vertically upward has zero speed at its highest point, yet its acceleration at that instant is $g = 9.8\,\mathrm{m\,s^{-2}}$ downward.
- **Statement II is FALSE:** Speed is the magnitude of velocity ($|v|$). If $|v| = 0$, the velocity v must also be zero.
- **Statement III is TRUE:** In one-dimensional motion, constant non-zero speed means velocity cannot change magnitude and cannot reverse direction smoothly without passing through zero speed; hence velocity is constant and $a = 0$.
- **Statement IV is FALSE:** A particle speeds up only when velocity and acceleration have the *same* sign ($v \cdot a > 0$). If the particle is moving in the negative x-direction ($v < 0$) with positive acceleration ($a > 0$), it is actually *slowing down*!

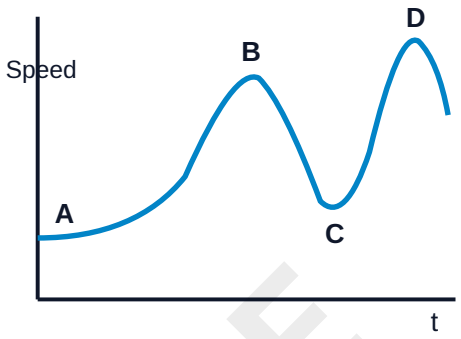
BOSONANCE

Practice Question 10 [Graphical Analysis — Comparing Equal Time Intervals on x-t and Speed-t Plots]

Consider the two graphs shown below. In Tutorial Diagram 5a, the position-time (x-t) plot of a particle in 1D motion is shown over three equal small time intervals 1, 2, and 3. In Tutorial Diagram 5b, the speed-time plot of a particle moving along a constant positive direction is shown with points $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$. Which of the following conclusions is **CORRECT**?



Tutorial Diagram 5a: x-t plot with equal intervals 1, 2, 3.



Tutorial Diagram 5b: Speed-t plot along constant direction.

(A) In Diagram 5a, average speed is greatest in interval 3, and average velocity is positive in intervals 1 and 2 and negative in interval 3; in Diagram 5b, instantaneous acceleration is zero at points $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and D.

(B) In Diagram 5a, average speed is greatest in interval 2 because x is maximum there; in Diagram 5b, acceleration is maximum at D.

(C) In Diagram 5a, average velocity is negative in interval 3 because x is negative there; in Diagram 5b, acceleration is non-zero at B and D.

(D) In Diagram 5a, average speed is least in interval 1; in Diagram 5b, velocity reverses sign at C.

Correct Answer: (A)
Worked Solution:

- In **Tutorial Diagram 5a**, the magnitude of slope $|\Delta x/\Delta t|$ is steepest in interval 3 and flattest (least) in interval 2 (near the peak). In intervals 1 and 2, x increases with t (positive slope $\implies \bar{v} > 0$), whereas in interval 3, x decreases with t (negative slope $\implies \bar{v} < 0$).
- In **Tutorial Diagram 5b**, the particle moves along a constant direction, so speed equals velocity. At points A, B (local peak), C (local trough), and D (local peak), the tangent to the speed-time curve is horizontal (slope $= 0$), meaning the instantaneous acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t} = 0$ at all four points $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and D.

BOSONANCE

Practice Question 11 [Assertion–Reason — Oscilliatory x-t Graph Signs]

For a particle executing one-dimensional simple harmonic motion along the x-axis with time period $T = 2.0\,\text{s}$, its position-time (x-t) graph is sinusoidal: starting at $x = 0$ at $t = 0$, moving to negative x for $0 < t < 1.0\,\text{s}$ (reaching minimum at $t = 0.5\,\text{s}$), and moving to positive x for $1.0\,\text{s} < t < 2.0\,\text{s}$ (reaching maximum at $t = 1.5\,\text{s}$).

Assertion (A): At $t = 0.3\,\text{s}$, the position x is negative, the velocity v is negative, and the acceleration a is positive; whereas at $t = 1.2\,\text{s}$, x is positive, v is positive, and a is negative.

Reason (R): On an x-t graph, the sign of velocity is given by the sign of the slope of the tangent ($\frac{\text{d}x}{\text{d}t}$), and the sign of acceleration is given by the concavity of the curve ($\frac{\text{d}^2x}{\text{d}t^2} > 0$ if curved upward, $\frac{\text{d}^2x}{\text{d}t^2} < 0$ if curved downward).

(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Correct Answer: (A)
Worked Solution:
At $t = 0.3\,\text{s}$ (between 0 and $0.5\,\text{s}$): the curve lies below the t-axis ($x < 0$), slopes downward ($v = \frac{\text{d}x}{\text{d}t} < 0$), and is concave upward near the trough ($a = \frac{\text{d}^2x}{\text{d}t^2} > 0$; also in SHM, $a = -\omega^2 x > 0$ when $x < 0$).
At $t = 1.2\,\text{s}$ (between $1.0\,\text{s}$ and $1.5\,\text{s}$): the curve lies above the t-axis ($x > 0$), slopes upward ($v > 0$), and is concave downward near the crest ($a < 0$). Thus both A and R are true, and R directly explains A.

BOSONANCE

Practice Question 12 [Matrix-Match — Physical Situations and Kinematics Graphs]

Match the graphical descriptions in **Column I** with the most appropriate physical situation in **Column II**:

Column I (Graph Description)	Column II (Physical Context)
(P) An x-t graph that is zero ($x = 0$) initially, rises linearly to a peak, then decreases linearly with a gentler slope to a negative constant value.	**(1)** A ball thrown vertically upward or dropped on a floor and rebounding repeatedly with reduced speed at each collision.
(Q) A v-t graph consisting of parallel downward-sloping straight segments that jump from negative velocity to a smaller positive velocity at each floor impact.	**(2)** A cricket bat striking a moving ball, or a hammer hitting a nail (large force acting for a very short duration).
(R) An a-t graph that is zero everywhere except for a single sharp, smooth positive pulse over a very short time interval.	**(3)** A ball resting on a smooth floor is kicked, rebounds from a wall with reduced speed, and moves past the kicker to stop at an opposite wall.

(A) $\mathrm{P} \to 3,\ \mathrm{Q} \to 1,\ \mathrm{R} \to 2$

(B) $\mathrm{P} \to 1,\ \mathrm{Q} \to 3,\ \mathrm{R} \to 2$

(C) $\mathrm{P} \to 3,\ \mathrm{Q} \to 2,\ \mathrm{R} \to 1$

(D) $\mathrm{P} \to 2,\ \mathrm{Q} \to 1,\ \mathrm{R} \to 3$

Correct Answer: (A)
Worked Solution:

- **(P) matches (3):** A ball at rest ($x = 0$) is kicked (moves with constant positive velocity to the wall), rebounds with lower speed (negative slope of smaller magnitude), and stops at the opposite wall (x becomes constant).
- **(Q) matches (1):** A bouncing ball under free fall has constant downward acceleration ($a = -g$, parallel negative slopes on v-t), and at each floor bounce its velocity reverses from downward to upward with reduced magnitude.
- **(R) matches (2):** A bat striking a ball or a hammer hitting a nail produces a brief, intense acceleration pulse over a short contact time while $a = 0$ before and after contact.

BOSONANCE

Practice Question 13 [Repeated Inelastic Bouncing Under Gravity]

A ball is dropped from rest from a height of $90\,\mathrm{m}$ onto a horizontal floor (take $g = 10\,\mathrm{m\,s^{-2}}$). At each collision with the floor, the ball loses one-tenth ($\frac{1}{10}$) of its speed. What is the speed of the ball just *after* its first collision with the floor, and at what time t after being dropped does it reach the highest point of its first rebound?

(A) $27\sqrt{2}\,\mathrm{m\,s^{-1}}$ and $5.7\sqrt{2}\,\mathrm{s}$

(B) $30\sqrt{2}\,\mathrm{m\,s^{-1}}$ and $6.0\sqrt{2}\,\mathrm{s}$

(C) $27\sqrt{2}\,\mathrm{m\,s^{-1}}$ and $3.0\sqrt{2}\,\mathrm{s}$

(D) $24\sqrt{2}\,\mathrm{m\,s^{-1}}$ and $5.4\sqrt{2}\,\mathrm{s}$

Correct Answer: (A)
Worked Solution:
1. Time to fall $h = 90\,\mathrm{m}$ from rest:

$$t_1 = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 90}{10}} = \sqrt{18} = 3\sqrt{2}\,\mathrm{s}$$

2. Speed just before the first collision:

$$v_1 = g t_1 = 10 \times 3\sqrt{2} = 30\sqrt{2}\,\mathrm{m\,s^{-1}}$$

3. Since the ball loses $\frac{1}{10}$ of its speed on impact, its upward rebound speed is $\frac{9}{10} v_1$:

$$v_{\text{rebound}} = 0.9 \times 30\sqrt{2} = 27\sqrt{2}\,\mathrm{m\,s^{-1}}$$

4. Time to rise from the floor to the peak of the first rebound:

$$t_2 = \frac{v_{\text{rebound}}}{g} = \frac{27\sqrt{2}}{10} = 2.7\sqrt{2}\,\mathrm{s}$$

Total time from release to the first rebound peak $= t_1 + t_2 = 3\sqrt{2} + 2.7\sqrt{2} = 5.7\sqrt{2}\,\mathrm{s}$ (or using $g = 9.8\,\mathrm{m\,s^{-2}}$, $t_1 = \frac{30}{7}\,\mathrm{s} \approx 4.29\,\mathrm{s}$ and $t_1 + t_2 = 1.9 \times 4.29 \approx 8.14\,\mathrm{s}$).

BOSONANCE

Practice Question 14 [Relative Motion — Overtaking of Two Trains]

Two trains A and B, each of length $200\,\mathrm{m}$, are moving on parallel tracks in the **same direction** with uniform speeds of $30\,\mathrm{m\,s^{-1}}$ and $20\,\mathrm{m\,s^{-1}}$, respectively. Initially, the front of train A is just behind the rear of train B. How much time does train A take to completely overtake train B?

(A) $20\,\mathrm{s}$

(B) $40\,\mathrm{s}$

(C) $8\,\mathrm{s}$

(D) $10\,\mathrm{s}$

Correct Answer: (B)
Worked Solution:
Relative velocity of train A with respect to train B:

$$v_{\mathrm{AB}} = v_{\mathrm{A}} - v_{\mathrm{B}} = 30 - 20 = 10\,\mathrm{m\,s^{-1}}$$

To completely overtake train B from rear-to-front alignment, train A must cover a relative distance equal to the sum of the lengths of both trains:

$$d_{\mathrm{rel}} = L_{\mathrm{A}} + L_{\mathrm{B}} = 200\,\mathrm{m} + 200\,\mathrm{m} = 400\,\mathrm{m}$$

Time required $= \frac{d_{\mathrm{rel}}}{v_{\mathrm{AB}}} = \frac{400\,\mathrm{m}}{10\,\mathrm{m\,s^{-1}}} = 40\,\mathrm{s}$.

6. Points to Ponder, Common Pitfalls & Final Revision Recap

Essential Points to Ponder (High-Yield NEET Traps)

1. **Choice of Origin and Axis Direction:** The origin and the positive direction of an axis are a matter of choice. Always specify this choice first before assigning algebraic signs ($+$ or $-$) to position, displacement, velocity, and acceleration.
2. **Speeding Up vs. Slowing Down:** If a particle is speeding up, its acceleration is in the direction of its velocity (v and a have the **same sign**, $v \cdot a > 0$). If its speed is decreasing, its acceleration is in the direction opposite to its velocity (v and a have **opposite signs**, $v \cdot a < 0$). This physical truth is completely independent of the choice of origin and axis!
3. **Sign of Acceleration Alone Does Not Indicate Speeding Up:** A negative acceleration ($a < 0$) does *not* necessarily mean retardation (slowing down). For example, if upward is chosen as $+y$, gravity gives $a = -g < 0$. For a ball thrown upward ($v > 0$), this negative acceleration decreases speed; for a ball falling downward ($v < 0$), the exact same negative acceleration increases speed!
4. **Zero Velocity Does Not Imply Zero Acceleration:** A particle may be momentarily at rest ($v = 0$) and yet have a non-zero acceleration ($a \neq 0$), such as a vertically projected ball at the highest point of its path ($v = 0, a = -g$).
5. **Exact Definitions vs. Constant-Acceleration Equations:** The calculus definitions $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ and $a = \frac{\mathrm{d}v}{\mathrm{d}t}$ are exact and universally valid for all motion. In contrast, the kinematic equations ($v = v_0 + at$, $x = x_0 + v_0 t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2a(x - x_0)$, and $\bar{v} = \frac{v_0 + v}{2}$) are valid **only when acceleration is constant in both magnitude and direction**.

BOSONANCE

Concise Formula Checklist for Quick Revision

- **Average Velocity & Speed:** $\bar{v} = \frac{\Delta x}{\Delta t}, \quad v_{\text{avg}} = \frac{\text{Total Path Length}}{\Delta t} \geq |\bar{v}|.$
- **Instantaneous Velocity & Speed:** $v = \frac{\mathrm{d}x}{\mathrm{d}t} = \text{slope of } x\text{-}t \text{ graph}, \quad \text{Instantaneous speed} = |v|.$
- **Instantaneous Acceleration:** $a = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}x} = \text{slope of } v\text{-}t \text{ graph}.$
- **Area under v-t Graph:** $\Delta x = \int_{t_1}^{t_2} v\,\mathrm{d}t$ (algebraic area = displacement; sum of absolute areas = distance).
- **Stopping Distance & Reaction Time:** $d_s = \frac{-v_0^2}{2a} \propto v_0^2; \quad t_r = \sqrt{\frac{2d}{g}}.$
- **Galileo's Law of Odd Numbers (from rest):** $s_1 : s_2 : s_3 : s_4 : \cdots = 1 : 3 : 5 : 7 : \ldots$
- **1D Relative Velocity:** $v_{\text{AB}} = v_{\text{A}} - v_{\text{B}} = -v_{\text{BA}}.$

BOSONANCE